

MOUNT KENYA UNIVERSITY
SCHOOL OF EDUCATION
DEPARTMENT OF CURRICULUM AND INSTRUCTION

**TEACHERS' COMPETENCE AND ITS INFLUENCE ON THE EFFECTIVE
IMPLEMENTATION OF THE COMPETENCY-BASED CURRICULUM IN
PUBLIC PRIMARY SCHOOLS IN HABASWEIN SUB-COUNTY, WAJIR
COUNTY, KENYA**

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A Research Proposal Submitted to the School of Education in Partial Fulfilment of the
Requirements for the Award of the Degree of Master of Education (Curriculum
Studies) of Mount Kenya University

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DECLARATION

This research proposal is my original work and has not been presented for the award of a degree or for any other award in any other university.

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Supervisors' Approval

This research proposal has been submitted for examination with our approval as the university supervisors.

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DEDICATION

I dedicate this work to my family, whose patience, prayers and encouragement have carried me through my postgraduate studies, and to the teachers of Habaswein Sub-County, who continue to serve their learners with dedication under some of the most demanding conditions in the country.

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ABBREVIATIONS AND ACRONYMS

APHRC	African Population and Health Research Center
ASAL	Arid and Semi-Arid Lands
BOM	Board of Management
CBA	Competency-Based Assessment
CBAM	Concerns-Based Adoption Model
CBC	Competency-Based Curriculum
CBE	Competency-Based Education
CIDP	County Integrated Development Plan
CPD	Continuous Professional Development
CSO	Curriculum Support Officer
CVI	Content Validity Index
ICT	Information and Communication Technology
KICD	Kenya Institute of Curriculum Development
KNEC	Kenya National Examinations Council
MKU	Mount Kenya University
MoE	Ministry of Education
NACOSTI	National Commission for Science, Technology and Innovation
PCIs	Pertinent and Contemporary Issues
PWPER	Presidential Working Party on Education Reform
SPSS	Statistical Package for the Social Sciences
TPACK	Technological Pedagogical Content Knowledge
TSC	Teachers Service Commission
UNESCO	United Nations Educational, Scientific and Cultural Organization

ABSTRACT

The Competency-Based Curriculum (CBC) was introduced in Kenya in 2017 with the aim of producing learners who can think critically, solve problems, communicate effectively and apply knowledge in real life. The success of the curriculum depends largely on teachers, who are expected to understand its aims, use learner-centred methods and assess learners using a variety of authentic tools. Studies in different parts of the country, however, show that many teachers have not yet acquired these competences, and the gap is likely to be wider in marginalised arid and semi-arid counties such as Wajir. The purpose of this study is to assess teachers' competence and its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County, Kenya. The study has four objectives, namely to assess teachers' knowledge and understanding of the principles and practices of the CBC; to analyse their pedagogical competence in implementing it; to identify the factors that affect their competence; and to establish the relationship between teachers' competence and effective implementation. The study is based on the Concerns-Based Adoption Model, the Teacher Professional Competence Model and the Technological Pedagogical Content Knowledge framework. A convergent parallel mixed methods design combining a descriptive survey and a correlational approach will be used. The target population is about 42 public primary schools, 42 head teachers, 336 teachers and 3 Curriculum Support Officers. Stratified random sampling will be used to select 21 schools, while a sample of 183 teachers, determined using a finite population formula, will be distributed proportionately across the strata. The head teachers of the 21 schools and all 3 Curriculum Support Officers will also take part. Data will be collected through questionnaires, semi-structured interviews and classroom observation. The instruments will be piloted in Wajir South Sub-County, with content validity checked by experts and reliability tested using Cronbach's alpha. Quantitative data will be analysed in SPSS using descriptive statistics, Pearson correlation and multiple linear regression, while qualitative data will be analysed thematically. The findings are expected to help the Ministry of Education, the Teachers Service Commission, the Kenya Institute of Curriculum Development and the County Government of Wajir to plan targeted support for teachers in marginalised areas.

Keywords: Teachers' competence, Competency-Based Curriculum, curriculum implementation, pedagogical competence, marginalised areas, Wajir County

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Education systems in many countries have in recent years moved away from content-heavy curricula and adopted curricula that focus on what learners are able to do with what they learn. Under this approach, commonly referred to as competency-based education, the emphasis is placed on skills, values and attitudes that a learner can apply in real situations, and less on the amount of content the learner can recall during an examination (Jentsch & König, 2022). This shift has coincided with a global shortage of teachers. UNESCO and the International Task Force on Teachers for Education 2030 (2024) estimate that 44 million additional primary and secondary school teachers will be needed worldwide by 2030, and that sub-Saharan Africa alone will require about 15 million of them. The report notes that recruiting new teachers will not be enough on its own, since teachers who are already in service also need support to acquire the competences that new curricula demand.

Teacher competence goes beyond possessing a teaching qualification. Jentsch and König (2022) describe it as a combination of professional knowledge, practical skills, beliefs and motivation which enables a teacher to deal with the day-to-day demands of the classroom. Drawing on international large-scale studies, the authors show that these attributes develop gradually, first during initial teacher training and later through professional development, and that they have a bearing on the quality of instruction that learners receive. In a related argument, Mishra et al. (2023) observe that teachers' knowledge has to keep changing as curricula and technologies change, because teachers are expected to bring together knowledge of content, of pedagogy and of the tools available to them. For this reason, the success of any curriculum reform depends to a large extent on the teachers who implement it.

A number of African countries have adopted competence-based curricula within the last decade. Wawire et al. (2026) carried out a scoping review of studies from Ghana, Kenya, Rwanda, South Africa, Tanzania, Uganda and Zambia and reported that, although policy documents in these countries promote learner-centred teaching and formative assessment, classroom practice in many schools has remained teacher-centred. In Tanzania, Kasuga and Kalolo (2025) identified several dilemmas affecting implementation in Njombe Region, among them inadequate preparation of teachers, uneven distribution of resources between schools and the continued attachment of society to examination results. In Uganda, Kidega et al. (2024) found that teachers in Gulu City were constrained by large classes, inadequate materials and insufficient

training, while Barasa et al. (2025) raised the concern that schools serving poorer communities were lagging behind in implementing the new lower secondary curriculum, which could widen inequality instead of reducing it.

In Kenya, the Competency-Based Curriculum (CBC) was introduced in 2017 to replace the 8-4-4 system. The rollout was phased, beginning with pre-primary and lower primary grades. The curriculum seeks to nurture the potential of every learner through seven core competencies, namely communication and collaboration, critical thinking and problem solving, imagination and creativity, citizenship, digital literacy, learning to learn, and self-efficacy (Akala, 2021). Under the new structure, a learner spends two years in pre-primary, six years in primary school, three years in junior school and three years in senior school before joining tertiary education. The government appointed the Presidential Working Party on Education Reform (PWPER) in 2022 to review the implementation of the reform. The Working Party recommended that the competency approach be retained, but it also recommended a reduction in the number of learning areas, a lighter assessment load and more investment in building the capacity of teachers (Presidential Working Party on Education Reform [PWPER], 2023). The reform has since been referred to more often as Competency-Based Education (CBE), although what teachers are expected to do in class has remained largely the same.

Assessment forms a major part of the reform. The Kenya National Examinations Council (KNEC, 2021) developed a competency-based assessment framework under which teachers are required to use formative tools such as observation schedules, rating scales, checklists, portfolios, projects and rubrics alongside summative tests. The framework assumes that teachers are able to design authentic tasks, judge learners' work against set criteria and keep proper records for each learner. Macheso, Kati, and Wafula (2024) found that several years after the rollout, many primary school teachers in western Kenya were still depending mostly on written tests, and that portfolios, anecdotal records and projects were used much less often. This suggests that many teachers are yet to acquire the competences on which the curriculum depends.

Studies conducted in other parts of Kenya have reported similar gaps. In Nairobi, Nyamira, Kirinyaga, Kinangop, Moiben and Bomet, researchers found that teachers had either not been trained for the new curriculum or had been trained within a short period that left gaps in their understanding (Isaboke et al., 2021; Karua et al., 2025; Keter & Wabuke, 2025; Mohamed et al., 2022; Mutai et al., 2024; Okeyo & Mokua, 2023). Mwangi et al. (2026) reported that 95% of the teachers in their study had attended CBC training, yet 65% of them still held negative perceptions of the curriculum. Attendance at training therefore does not necessarily mean that a teacher is competent or committed to the curriculum. Mwang'ombe (2021) had earlier described teachers' understanding

of the CBC as vague and noted that this affected their planning, teaching and assessment. An evaluation by the African Population and Health Research Center (APHRC, 2022) also found that although some teachers were delivering effective CBC lessons, many lessons fell short in learner engagement and in the integration of core competencies.

The challenges mentioned above are likely to be greater in the arid and semi-arid lands (ASAL) of northern Kenya. Hussein (2023) reported that CBC implementation in the region was hindered by poor infrastructure, a shortage of learning resources and limited training of teachers. Ongubo (2025) found that more than 60% of public primary school teachers in Township Ward, Garissa County, had not received comprehensive in-service training on the CBC. Wajir County, where the present study will be carried out, faces many of the same problems. The County Integrated Development Plan lists a shortage of teachers, long distances to school, recurrent drought that affects enrolment and attendance, and a pastoralist lifestyle that makes it difficult for children to attend school continuously (County Government of Wajir, 2023). Insecurity in parts of the region has also led many teachers from outside the county to seek transfers, and schools have increasingly relied on teachers employed by Boards of Management.

Habaswein Sub-County is located in the southern part of Wajir County along the Garissa–Wajir road. Most public primary schools in the sub-county serve pastoralist and agro-pastoralist communities, and some of them operate low-cost boarding facilities to keep children from mobile households in school. Teachers in these schools often handle large classes with few teaching materials, and electricity and internet connectivity are unreliable. Opportunities for professional development are also limited. Despite these conditions, the teachers implement the same curriculum as their colleagues in well-equipped urban schools, and their learners sit the same national assessments. There is need, therefore, to establish the level of competence of teachers in Habaswein, the factors that influence it and how it relates to the implementation of the CBC in their classrooms.

1.2 Statement of the Problem

Effective implementation of the CBC requires teachers who understand the aims of the curriculum, use learner-centred methods, integrate core competencies and pertinent and contemporary issues in their lessons and assess learners through authentic tasks. Ideally, all public primary school teachers in Kenya should have these competences, so that a learner in Habaswein receives an education comparable to that of a learner in Nairobi or Nakuru. The government has taken steps towards this through national training programmes, the supply of curriculum designs and textbooks, and the deployment of Curriculum Support Officers (PWPER, 2023).

However, available studies indicate that there is still a wide gap between what the curriculum expects of teachers and what many teachers are able to do. Many teachers have limited understanding of CBC principles, find it difficult to prepare schemes of work and lesson plans in the required format and use only a few assessment tools (Karua et al., 2025; Macheso, Kati, & Wafula, 2024; Mwang'ombe, 2021). In northern Kenya the situation is made worse by teacher shortages, insecurity, poor infrastructure and limited in-service training (Hussein, 2023; Ongubo, 2025). Where teachers lack the required competence, implementation is likely to be superficial, with schools adopting the terminology of the new curriculum while teaching remains teacher-centred and examination-oriented (Awiti, 2023; Wawire et al., 2026). Learners in marginalised areas may then fail to acquire the competences that the reform was intended to develop, and existing inequalities may deepen (Barasa et al., 2025).

Most of the studies on teacher preparedness for the CBC have been conducted in urban or relatively well-served counties such as Nairobi, Machakos, Kirinyaga, Nyandarua and Uasin Gishu (Isaboke et al., 2021; Mogere, 2022; Mutua & Waweru, 2023). Few studies have been done in Wajir County, and the researcher is not aware of any study that has examined teachers' competence and its influence on CBC implementation in Habaswein Sub-County. In addition, many of the earlier studies measured preparedness mainly by asking whether teachers had attended training. They did not assess teachers' actual knowledge, classroom practice and assessment skills, and few of them tested statistically how these relate to the effectiveness of implementation. This lack of context-specific evidence makes it difficult for the Ministry of Education, the Teachers Service Commission and the County Government of Wajir to plan targeted support for teachers. It is against this background that the present study seeks to assess teachers' competence and its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County.

1.3 Purpose of the Study

The purpose of this study is to assess teachers' competence and establish its influence on the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya, in order to generate evidence that can guide teacher support and policy in marginalised areas.

1.4 Research Objectives

1.4.1 General Objective

The general objective of the study is to assess teachers' competence and its influence on the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya.

1.4.2 Specific Objectives

The study will be guided by the following specific objectives:

- i. To assess the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- ii. To analyse the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iii. To identify the factors affecting teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iv. To establish the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.

1.5 Research Questions

The study will seek to answer the following research questions:

- i. What is the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- ii. What is the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?

- iii. What factors affect teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- iv. What is the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?

1.6 Research Hypotheses

The following null hypotheses will be tested at 0.05 level of significance in relation to the fourth objective:

- H01: There is no statistically significant relationship between teachers' knowledge and understanding of the CBC and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.
- H02: There is no statistically significant relationship between teachers' pedagogical competence and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.
- H03: Teachers' overall competence has no statistically significant influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County.

1.7 Justification of the Study

The CBC is one of the biggest education reforms that Kenya has undertaken, and the first learners to go through it have only recently joined junior and senior school. Information on how the curriculum is being implemented in different regions is therefore timely. The reform was also meant to promote equity. Studies from Uganda and Tanzania, however, show that competence-based reforms may increase the gap between advantaged and disadvantaged schools where teachers do not receive equal support (Barasa et al., 2025; Kasuga & Kalolo, 2025). Habaswein is one of the areas where such a gap would be expected.

The PWPER (2023) recommended more investment in teacher capacity building, and the Teachers Service Commission continues to organise in-service training for teachers. Such investment is more likely to succeed if it is based on evidence about the areas in which teachers are competent, the areas in which they struggle and the factors that hold them back. Lastly, most Kenyan studies on the CBC have been done in central, western and urban parts of the country. A study in the North Eastern region will add to the national body of evidence, which currently says little about this part of Kenya.

1.8 Significance of the Study

The findings of the study may be of benefit to various stakeholders. The Ministry of Education and the Kenya Institute of Curriculum Development (KICD) may use them to improve curriculum support for ASAL counties, for example by simplifying curriculum materials or giving assessment guidelines suited to local conditions. The Teachers Service Commission (TSC) may use the findings when planning in-service training and professional development and when reviewing staffing and incentive policies for hardship areas.

Teacher training colleges and universities may find the findings useful in improving the pre-service training of teachers, particularly in learner-centred methods, competency-based assessment and teaching in multi-grade and poorly resourced classrooms. The County Government of Wajir may use the findings to plan its support for basic education and to lobby for more resources from the national government. Head teachers and Curriculum Support Officers in Habaswein may use them to organise school-based professional learning, mentoring and supervision, while teachers may be able to identify areas of their own practice that need improvement. The study will also add to the limited literature on CBC implementation in marginalised areas and may point to areas for further research.

1.9 Limitations of the Study

Some of the data will be self-reported by teachers through questionnaires, and some respondents may rate their competence higher than it actually is. To reduce this problem, the researcher will compare the questionnaire responses with data from classroom observation and from interviews with head teachers and Curriculum Support Officers.

Schools in the sub-county are far apart, the roads are poor and some areas experience insecurity from time to time, which may make some schools difficult to reach. The researcher will carry out fieldwork during the dry months, work with the local education and security offices and use research assistants who are familiar with the area. Some schools may also be closed for a period, or have low attendance, because of drought or the movement of pastoralist families. In such cases a replacement school will be selected from the same stratum. Since the study will cover only one sub-county, its findings may not apply to the whole country, although they are likely to be relevant to other ASAL areas with similar conditions.

1.10 Delimitations of the Study

The study will be confined to public primary schools in Habaswein Sub-County, Wajir County. Private primary schools, pre-primary centres and junior schools will not be included, since their staffing, funding and management are different and would make comparison difficult. The respondents will be TSC and Board of Management teachers teaching Grades 1 to 6, head teachers of the sampled schools and the Curriculum Support Officers serving the sub-county. The study will focus on teachers' knowledge and understanding of the CBC, their pedagogical competence, the factors affecting their competence and the relationship between competence and effective implementation. Other factors such as parental involvement and learner characteristics will be acknowledged but will not be studied in depth.

1.11 Assumptions of the Study

The study will be based on the following assumptions:

- i. That respondents will give honest and accurate information about their knowledge, practices and experiences in implementing the CBC.
- ii. That all public primary schools in Habaswein Sub-County are implementing the CBC in line with national guidelines issued by the Ministry of Education and KICD.
- iii. That teachers in the sampled schools have had at least some exposure to CBC training, whether through national rollout programmes, school-based sessions or self-directed learning.
- iv. That the security situation in the sub-county will remain stable enough to allow data collection within the planned period.

1.12 Operational Definition of Terms

Competency-Based Curriculum (CBC): The curriculum introduced in Kenya from 2017 which aims at developing learners' core competencies, values and practical skills through learner-centred teaching and continuous assessment. In this study the term also refers to Competency-Based Education (CBE), the name now used for the reform.

Teachers' competence: The knowledge, teaching skills, assessment skills and attitudes that enable a teacher to plan, teach and assess CBC lessons as required. In this study it will be measured through teachers' knowledge of the CBC, their pedagogical competence and their assessment practices.

Knowledge and understanding of CBC: The extent to which a teacher is able to explain the goals, core competencies, values, pertinent and contemporary issues, learning areas and assessment requirements of the CBC.

Pedagogical competence: The ability of a teacher to plan and deliver lessons using learner-centred methods, integrate core competencies and values in lessons, manage learner activities and use locally available resources.

Effective implementation of CBC: The extent to which classroom practice reflects the intentions of the CBC, as shown by learner participation, integration of core competencies, use of competency-based assessment, lesson planning in the prescribed format and coverage of learning areas within the set time.

Factors affecting competence: Conditions within the school, the teacher and the surrounding environment, such as training, workload, resources, supervision and the physical environment, which enhance or limit a teacher's ability to implement the CBC.

Marginalised area: An area with poor access to social services, poor infrastructure and limited economic opportunities, as is the case in many ASAL counties including Wajir.

1.13 Organisation of the Proposal

This proposal is organised in three chapters. Chapter One covers the background to the study, the statement of the problem, the purpose, objectives, research questions and hypotheses, the justification and significance of the study, its limitations, delimitations and assumptions, and the operational definition of terms. Chapter Two reviews the theoretical and empirical literature related to the objectives of the study, presents the conceptual framework and highlights the research gaps. Chapter Three describes the research methodology, including the research design, location, target population, sampling techniques and sample size, research instruments, pilot study, data collection and analysis procedures, and ethical considerations. The references and appendices are presented at the end.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature related to the study. The first part presents the theories on which the study is based. This is followed by an overview of the Competency-Based Curriculum in Kenya and a review of empirical studies under each of the four objectives. The chapter then gives a critique of the literature, the conceptual framework and a summary of the research gaps.

2.2 Theoretical Review

The study is guided by three theories: the Concerns-Based Adoption Model, the Teacher Professional Competence Model and the Technological Pedagogical Content Knowledge framework. The first explains how teachers adopt a new curriculum over time. The second explains what teacher competence consists of and how it affects classroom practice, while the third explains the kind of integrated knowledge a teacher needs to teach content effectively with the tools available. The third theory is relevant because digital literacy is one of the core competencies of the CBC.

2.2.1 Concerns-Based Adoption Model

The Concerns-Based Adoption Model (CBAM) was developed in the 1970s by Gene Hall and colleagues at the University of Texas. It was meant to explain how individuals, especially teachers, experience and adopt innovations in education. The model is based on the view that change is a process and not a one-off event, and that change takes place in individuals before it becomes part of an institution. It has three dimensions, which are stages of concern, levels of use and innovation configurations (Okeyo & Mokuu, 2023).

The stages of concern describe how the feelings of a teacher towards an innovation change with time. At the beginning, teachers are mainly concerned about getting information on the innovation and about how it will affect them personally. When they start using it, they become more concerned with management issues such as time, materials and organisation. At later stages, their concern moves to the effect of the innovation on learners, working with colleagues and improving the innovation. The levels of use describe what teachers actually do with the innovation. They range from non-use, to mechanical and routine use, and finally to refinement and renewal. Innovation configurations refer to the different ways in which different teachers put the same innovation into practice (Okeyo & Mokuu, 2023).

CBAM has been used in several studies on curriculum change in Africa. In Kenya, Okeyo and Mokua (2023) used it in their study of teachers' preparedness for the CBC in Nyamira South Sub-County and found that most teachers were ill-prepared to implement the curriculum. In the present study, the model will help in interpreting teachers' knowledge and practice as stages in a process of growth. A teacher with little understanding of the CBC is likely to be at an early stage of concern and to use the curriculum mechanically. A competent teacher is more likely to be concerned about learners' progress and to adapt the curriculum to suit them. The model also suggests that teachers need support that responds to their concerns, which relates to the third objective of the study.

The model has, however, been criticised for focusing on the individual teacher and giving little attention to structural problems such as funding, class size, infrastructure and policy, which can hinder implementation regardless of the teacher's level of concern. It has also been criticised for suggesting that teachers move through the stages in a straight line, whereas teachers may go back to earlier concerns when conditions change. For this reason, the study combines CBAM with a model of teacher competence and gives attention to the conditions found in Habaswein.

2.2.2 Teacher Professional Competence Model

The Teacher Professional Competence Model is based on the work of Sigrid Blömeke and other researchers who argued that competence should be seen as a continuum. According to the model, a teacher's dispositions, which include professional knowledge, beliefs, motivation and self-regulation, influence situation-specific skills. These skills are the ability to notice what is happening in class, interpret it and decide how to respond. The skills in turn lead to observable performance in the classroom (Jentsch & König, 2022).

Jentsch and König (2022) explain that the cognitive part of competence is made up of content knowledge, pedagogical content knowledge and general pedagogical knowledge. The affective-motivational part is made up of teachers' beliefs about teaching and learning, their enthusiasm and their self-efficacy. Based on international large-scale studies, the authors show that both parts affect the quality of instruction and the achievement of learners, and that both can be developed during initial teacher education and through professional development. One implication of the model is that knowledge alone does not make a good teacher. A teacher may know the principles of the CBC and still fail to apply them because of low confidence, negative beliefs about the curriculum or difficulty in responding to classroom situations.

The model is relevant to the present study in three ways. It provides a structure for the independent variable, which the study divides into knowledge and understanding of the CBC and pedagogical competence. It also explains why attendance at training is not a good measure of competence, which is a weakness of many earlier studies on the CBC. In addition, since the model treats competence as something that can be seen in practice, it supports the use of classroom observation alongside questionnaires. The main weakness of the model is that it was developed mostly in countries with well-resourced school systems, and it may therefore not fully account for the way difficult conditions in marginalised areas limit what competent teachers can do. The study addresses this through the third objective.

2.2.3 Technological Pedagogical Content Knowledge Framework

The Technological Pedagogical Content Knowledge (TPACK) framework was proposed by Punya Mishra and Matthew Koehler in the mid-2000s as an extension of the idea of pedagogical content knowledge. The framework states that effective teaching requires teachers to combine content knowledge (what to teach), pedagogical knowledge (how to teach) and technological knowledge (the tools that support teaching). The areas where these forms of knowledge overlap, such as pedagogical content knowledge and technological pedagogical knowledge, represent the specialised knowledge that distinguishes an expert teacher (Mishra et al., 2023).

Mishra et al. (2023) revisited the framework and emphasised that teachers also need contextual knowledge, that is, knowledge of the conditions, limitations and opportunities of the setting in which they teach. This is relevant to Habaswein, where electricity, digital devices and internet access are not always available. Digital literacy is one of the core competencies of the CBC and teachers are expected to integrate it in all learning areas. Murithi and Yoo (2021), however, found that many Kenyan primary school teachers had only received basic computer literacy training and were not able to integrate technology meaningfully in CBC lessons. The TPACK framework will help the study to examine whether teachers are able to combine content, pedagogy and the technology available to them, including low-cost resources. Because the knowledge areas in TPACK overlap and are difficult to measure separately, the framework will be used to interpret the findings and not as a measurement tool.

2.3 The Competency-Based Curriculum in Kenya

Kenya adopted a competency-based curriculum partly because the 8-4-4 system had become too examination-oriented and was not producing school leavers with the skills and values required in a changing economy (Akala, 2021). The Basic Education Curriculum Framework set out the vision of nurturing every learner's potential. It

identified seven core competencies and a set of national values, together with pertinent and contemporary issues (PCIs) such as health, environmental awareness, citizenship and life skills, which are to be integrated across the learning areas. Learners are expected to discover, create and apply knowledge through activities, while the teacher acts as a facilitator rather than the only source of knowledge (Thukia, 2025).

Thukia (2025) analysed the reform using Basil Bernstein's ideas on curriculum and pedagogy. According to this analysis, the CBC is a move from a teacher-controlled model of teaching to a model in which learners have more control over the pace and order of learning. This places heavy demands on teachers, who must be confident enough to give learners some control while still guiding the learning process. Where teachers lack this confidence or the necessary skills, schools may adopt the terminology of the new curriculum while old classroom routines continue.

Assessment is a key part of the CBC. Under the KNEC (2021) framework, teachers carry out school-based assessment using a variety of tools and record learners' progress at four levels: exceeding expectations, meeting expectations, approaching expectations and below expectations. The PWPER (2023) noted that implementation of the curriculum had been affected by inadequate preparation of teachers, an overloaded timetable and a shortage of resources. It recommended a reduction in the number of learning areas and more investment in teacher capacity building. These recommendations show that teacher competence is regarded as critical to the success of the reform. They also show that the curriculum is still being reviewed, and teachers will need to keep updating their knowledge.

2.4 Empirical Review

This section reviews studies related to each of the four objectives of the study. Studies done outside Kenya are reviewed first where available, followed by those done in Kenya.

2.4.1 Teachers' Knowledge and Understanding of the Competency-Based Curriculum

Wawire et al. (2026) conducted a scoping review of literature on pedagogical and assessment practices in schools implementing competence-based curricula in Ghana, Kenya, Rwanda, South Africa, Tanzania, Uganda and Zambia. The review established that although the reforms call for a shift from content coverage to the development of competences, much of the classroom practice in these countries still reflects the older approach. The review, however, was a synthesis of existing studies and did not collect primary data on teachers' knowledge in any particular context.

Kasuga and Kalolo (2025) studied the dilemmas facing the implementation of the competency-based curriculum in Njombe Region, Tanzania. The study found that many teachers had not received adequate training and support, and it identified teachers' readiness as one of the main dilemmas affecting implementation. In Uganda, Kidega et al. (2024) examined the challenges facing teachers in implementing the competency-based curriculum in secondary schools in Gulu City. The study found that the challenges were linked to insufficient training, large classes and inadequate materials. Both studies were conducted at secondary school level and outside Kenya.

In Kenya, Mwang'ombe (2021) examined teachers' understanding and skills in relation to the CBC and concluded that their understanding of the curriculum was vague and uneven. Isaboke et al. (2021) studied teacher preparedness and the implementation of the CBC in public pre-primary schools in Nairobi City County. The study targeted 450 pre-primary teachers, 225 ECD centre managers and 225 head teachers. It found that 65.9% of the teachers had not received any CBC training and that implementation was hindered by lack of training, inadequate facilities and large classes. The study was done in pre-primary schools in an urban county, and its findings may not apply to primary schools in ASAL areas.

Mutua and Waweru (2023) investigated teachers' competence and its influence on the implementation of the CBC in public primary schools in Machakos County. The study used a descriptive survey design and targeted 79 public primary schools, 941 teachers, 79 deputy head teachers and 79 head teachers. It found that yearly in-service training for teachers was adequate, but implementation was hindered by lack of enough teaching and learning materials and a high pupil-teacher ratio. Karua et al. (2025) studied teacher preparedness to implement the CBC in public primary schools in Kinangop Sub-County. The study targeted 268 lower primary teachers in 46 schools and found that the majority of teachers (52.3%) were not competent in preparing schemes of work, and that teachers had low capacity to select appropriate teaching and learning approaches. This shows that awareness of the curriculum does not always mean that a teacher knows how to plan for it.

Mwangi et al. (2026) studied teacher training and perceptions of the CBC in Kenyan schools. They found that although 95% of the teachers had been trained to implement the curriculum, 65% held negative perceptions of it. This finding is in line with the argument by Jentsch and König (2022) that competence includes beliefs and motivation as well as knowledge. Keter and Wabuke (2025) investigated teacher preparedness and the implementation of the CBC in public junior secondary schools in Bomet County and found that teacher preparedness significantly influenced implementation.

Few studies have been done in the ASAL counties. Hussein (2023) examined the challenges facing CBC implementation in northern Kenya and found that teachers had limited exposure to training and that poor infrastructure and inadequate learning resources made proper grading difficult. Ongubo (2025) assessed public primary school teacher preparedness for CBC adoption in Township Ward, Garissa County, and found that more than 60% of the teachers had not received comprehensive in-service training. Neither of the two studies measured teachers' knowledge using a knowledge scale or tested its relationship with implementation. The present study intends to fill this gap in Habaswein Sub-County.

2.4.2 Teachers' Pedagogical Competence in Implementing the CBC

Pedagogical competence refers to what a teacher does in class, including how the teacher plans, organises learning activities, involves learners and uses resources. The CBC requires teachers to use learner-centred methods such as inquiry, cooperative learning, role play, projects, field visits and problem solving, and to integrate values and PCIs in their lessons. In their review, Wawire et al. (2026) found that classroom practice in African countries implementing competence-based curricula has remained largely teacher-centred, with learners given few opportunities to work together, investigate or create. This was attributed to factors such as large classes, a shortage of materials and inadequate preparation of teachers.

In Kenya, APHRC (2022) evaluated the quality of CBC implementation by examining lessons in schools with the aim of identifying the features of effective lessons, such as clear learning outcomes, active participation of learners and deliberate integration of core competencies. The evaluation found that effective CBC lessons were being delivered in some public schools, but that the quality of lessons varied widely. Mohamed et al. (2022) studied the pedagogical preparedness of geography teachers for the implementation of the CBC in public secondary schools in Kirinyaga County. The study targeted 24 schools, 24 head teachers and 100 geography teachers, from which a sample of 40 respondents was drawn. It found that teachers' pedagogical preparedness for the CBC was low. The small sample, however, limits the extent to which the findings can be generalised.

Alwang'a and Were (2026) examined the status of teachers' pedagogical knowledge and the implementation of the CBC in public primary schools in Sabatia Sub-County, Vihiga County. Their study linked the quality of implementation to teachers' pedagogical knowledge. Mutai et al. (2024) studied CBC implementation and teacher preparedness in public schools in Moiben Sub-County, Uasin Gishu County, using a descriptive survey of 605 respondents, including teachers, head teachers and

Curriculum Support Officers. The study found that teacher preparedness had a significant influence on the implementation of the CBC.

Macheso, Kati, and Wafula (2024) investigated the extent to which teachers in western Kenya used competency-based assessment tools. They found that teachers used written tests much more than portfolios, projects and anecdotal records, although the KNEC (2021) framework requires the use of a variety of tools. In another study of Grade 6 science and technology teachers in Bungoma, Busia and Trans-Nzoia counties, Macheso, Wafula, and Kati (2024) found that teachers' self-efficacy was related to the extent to which they implemented competency-based assessment. Teachers who were confident about assessment implemented it more fully. These findings suggest that pedagogical competence depends on teachers' belief in their own ability as well as on their skills.

Murithi and Yoo (2021) studied teachers' use of ICT in implementing the CBC in Kenyan public primary schools. The study involved 351 teachers who responded to an online questionnaire. Most of the teachers considered ICT necessary for the CBC, but they had received only basic computer literacy training and had difficulty integrating technology in their lessons. The study was done online, and teachers in remote areas with poor connectivity may not have been well represented. It is not clear how teachers in areas such as Habaswein, where electricity and internet access are limited, handle the digital literacy competency.

2.4.3 Factors Affecting Teachers' Competence in Implementing the CBC

The literature identifies several factors that affect teachers' competence in implementing the CBC. These include training and professional development, teaching resources and infrastructure, workload and staffing, supervision and support, attitudes, and the wider context in which schools operate.

UNESCO and the International Task Force on Teachers for Education 2030 (2024) argue that professional development improves teaching only when it is continuous, practical and based on the needs of teachers. In Kenya, the initial CBC training was widely described as short and rushed. Okeyo and Mokua (2023) studied teacher preparedness for the CBC in public secondary schools in Nyamira South Sub-County using a convergent mixed methods design and a sample of 483 respondents. They found that the schools were ill-prepared and most teachers had not been trained, and they recommended that training should not be done hurriedly as a public relations exercise. Ongubo (2025) also found that the coverage of in-service training in Township Ward, Garissa County, was limited. Mogere (2022) studied the determinants of CBC

implementation in public primary schools in Nairobi County and identified teacher training as an important determinant.

Teaching resources and infrastructure also affect teachers' ability to implement the curriculum. Cheruiyot (2024) studied the challenges facing CBC implementation in junior schools in Kenya and identified a shortage of learning materials, inadequate classrooms and limited technological infrastructure. The study also found resistance to change among teachers, parents and learners. Hussein (2023) reported that poor infrastructure and lack of learning resources affected proper grading in northern Kenya, while Mutua and Waweru (2023) found that inadequate teaching and learning materials hindered implementation in Machakos County. Where resources are lacking, even competent teachers may not be able to use the methods they know to be effective.

Workload and staffing determine the time that teachers have for planning, assessment and reflection. The CBC requires detailed lesson plans and continuous assessment records for every learner, which is demanding when classes are large. Isaboke et al. (2021) listed large classes and understaffing among the challenges facing implementation in Nairobi, and Mutua and Waweru (2023) identified a high pupil-teacher ratio as a constraint. The shortage of teachers is more serious in ASAL counties. The County Government of Wajir (2023) acknowledges the shortage of teachers as a major problem in the county. Many schools therefore depend on Board of Management teachers, some of whom may not have received formal CBC training.

Supervision and support from head teachers and Curriculum Support Officers also influence teacher competence. Awiti (2023) examined factors affecting the implementation of the CBC in primary schools in Nairobi City and pointed out the importance of the support that teachers receive within their schools. Keter and Wabuke (2025) also emphasised the role of continuous support in improving teacher preparedness. It is expected that teachers become more confident where head teachers observe lessons regularly, check schemes of work and organise peer learning.

Teachers' attitudes are another factor. Mwangi et al. (2026) found that most teachers had negative perceptions of the CBC even after being trained, and Cheruiyot (2024) reported resistance to change among teachers. According to the Teacher Professional Competence Model, such attitudes influence how teachers interpret classroom situations and how they respond. A teacher who does not believe in learner-centred methods may go back to the lecture method, especially when under pressure to complete the syllabus.

The context in which schools operate is also important, especially in marginalised areas. Barasa et al. (2025) studied the implementation of the lower secondary competence-

based curriculum in Uganda and found that schools serving learners from poor backgrounds were less able to implement the reform than schools serving wealthier learners. They warned that the reform could increase inequality unless deliberate support was given. Kasuga and Kalolo (2025) reported that rural schools in Tanzania faced serious shortages of materials and infrastructure. In northern Kenya, drought, the movement of pastoralist families, long distances and insecurity affect learners' attendance and teachers' access to training (County Government of Wajir, 2023; Hussein, 2023). These conditions may lower teachers' morale and increase staff turnover. Few studies have examined how such conditions relate to teachers' competence in implementing the CBC in Kenya, and the present study will address this.

2.4.4 Teachers' Competence and Effective Implementation of the CBC

Several Kenyan studies have looked at the relationship between teacher-related factors and CBC implementation, although most of them used preparedness rather than competence as the independent variable. Mutai et al. (2024) found a significant positive relationship between teacher preparedness and CBC implementation in Moiben Sub-County. Keter and Wabuke (2025) found that teacher preparedness significantly influenced the implementation of the CBC in public junior secondary schools in Bomet County. Mutua and Waweru (2023) examined teachers' competence in terms of values, attitudes, knowledge and skills together with the extent of CBC implementation in Machakos County, and concluded that implementation depended on both the teachers and the conditions under which they worked, such as the availability of materials and the pupil-teacher ratio.

Alwang'a and Were (2026) focused on pedagogical knowledge as one aspect of competence and linked it to the implementation of the CBC in Sabatia Sub-County. Macheso, Wafula, and Kati (2024) found that teacher efficacy was related to the implementation of competency-based assessment. The evaluation by APHRC (2022) also suggests that the quality of CBC lessons depends a great deal on teachers' planning and their understanding of the competences that the lessons are meant to develop.

Outside Kenya, the review by Wawire et al. (2026) shows that the gap between competence-based policy and classroom practice is closely related to teacher preparation and the resources available in schools. Kidega et al. (2024) recommended continuous training and support for teachers in Uganda after finding that many of the challenges they faced were due to inadequate preparation for the new curriculum. Jentsch and König (2022) provide the theoretical basis for these findings, since they argue that teacher competence influences the quality of instruction, which in turn influences learners' outcomes.

These studies have three main weaknesses. Most of them relied on self-report questionnaires, which may overstate teachers' competence. Few of them separated the different aspects of competence and tested their individual contribution to implementation. In addition, very few were conducted in the ASAL counties of North Eastern Kenya. The present study will combine questionnaires with classroom observation and interviews, test the separate and combined influence of knowledge and pedagogical competence on implementation, and focus on Habaswein Sub-County.

2.5 Critique of Existing Literature

The literature reviewed agrees that teacher competence is important for the success of the CBC and that many teachers in Kenya have not yet acquired the competences required. The literature, however, has some weaknesses. First, most of the studies were done in Nairobi, central Kenya, the Rift Valley and western Kenya (Isaboke et al., 2021; Karua et al., 2025; Mohamed et al., 2022; Mutai et al., 2024). The few studies done in northern Kenya, such as Hussein (2023) and Ongubo (2025), were mainly descriptive and did not test the relationship between competence and implementation.

Second, the literature is not consistent in the use of concepts. Terms such as preparedness, readiness, competence and capacity are often used to mean the same thing, and many studies measured preparedness mainly by asking whether teachers had attended training. The findings of Mwangi et al. (2026) show that this is a weak measure, since trained teachers may still lack understanding or have negative attitudes towards the curriculum. The present study uses a clearer definition of competence based on the Teacher Professional Competence Model (Jentsch & König, 2022), and separates knowledge from pedagogical competence.

Third, many studies relied only on questionnaires, which are prone to social desirability bias. Only a few, such as APHRC (2022) and Okeyo and Mokua (2023), used observation or document analysis to confirm what teachers reported. Some studies also used small samples. Mohamed et al. (2022), for instance, based their findings on 40 respondents. The present study will use stratified random sampling, an adequate sample size and data from several sources.

Lastly, most studies treated the context of schools as background information rather than as a factor that interacts with teacher competence. Barasa et al. (2025) and Kasuga and Kalolo (2025) show that context affects how competence-based reforms are implemented, but Kenyan studies have rarely examined how drought, insecurity and pastoralist movement affect teachers' professional growth. The present study will examine these factors.

2.6 Conceptual Framework

The conceptual framework in Figure 2.1 shows the expected relationship between the independent, intervening and dependent variables of the study.

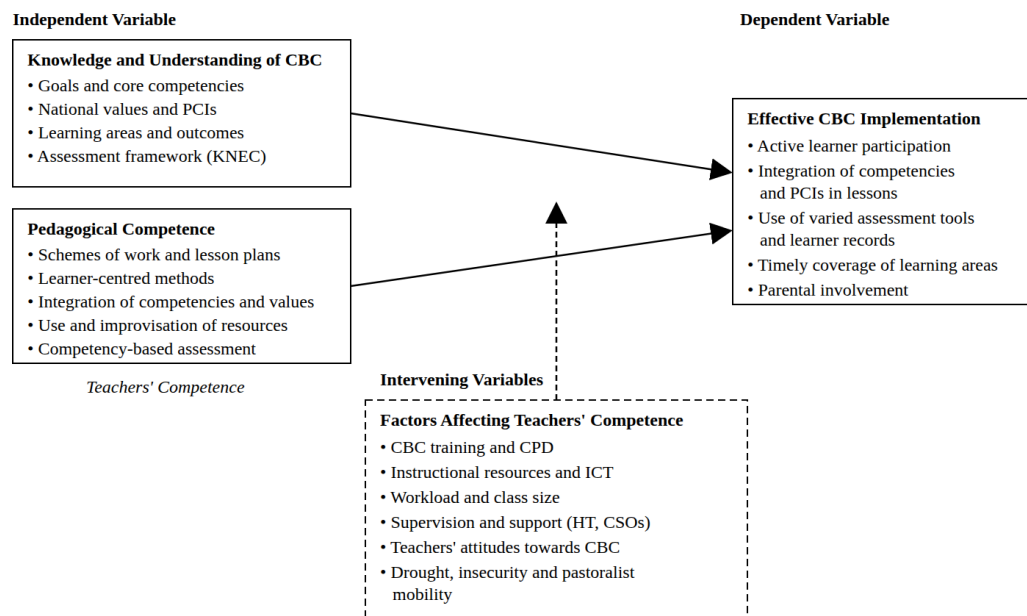


Figure 2.1: Conceptual Framework

Source: Researcher (2026)

The independent variable is teachers' competence, which has two dimensions. The first is knowledge and understanding of the CBC, which refers to teachers' understanding of the goals, core competencies, values, PCIs, learning areas and assessment requirements of the curriculum. The second is pedagogical competence, which refers to teachers' ability to prepare schemes of work and lesson plans in the required format, use learner-centred methods, integrate core competencies and values in lessons, use and improvise resources, and carry out competency-based assessment.

The dependent variable is the effective implementation of the CBC. It will be measured using indicators that reflect the intentions of the curriculum. These are active participation of learners, integration of core competencies and PCIs in lessons, use of a variety of assessment tools and proper learner records, coverage of learning areas within the set time, and involvement of parents in learning activities.

The factors affecting teachers' competence are the intervening variables. They include the adequacy of CBC training and professional development, availability of teaching resources and ICT facilities, teachers' workload and class size, supervision and support from head teachers and Curriculum Support Officers, teachers' attitudes towards the CBC, and local conditions such as drought, insecurity and the movement of pastoralist

families. These factors may strengthen or weaken the relationship between competence and implementation. For example, a teacher who understands the CBC well may still find it difficult to use group work in a class of 90 learners without desks. Government policy, including the reforms recommended by the PWPER (2023), is acknowledged as a wider influence, but it will not be measured in this study.

2.7 Summary of Literature and Research Gaps

The literature shows that the CBC requires teachers to have a good understanding of the curriculum, skills in learner-centred teaching and the ability to assess competences. CBAM explains that teachers adopt new ideas gradually and need support that addresses their concerns. The Teacher Professional Competence Model shows that competence is made up of knowledge, beliefs and situation-specific skills, while the TPACK framework points to the combined knowledge that teachers need in order to teach content effectively with the tools available in their setting. Studies in Kenya and in other African countries show that gaps in teacher competence are common and are influenced by training, resources, workload, supervision, attitudes and context. Table 2.1 summarises the main studies reviewed and the gaps identified.

Table 2.1: Summary of Research Gaps

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Isaboke et al. (2021)	Teacher preparedness and CBC implementation in pre-primary schools, Nairobi	65.9% of teachers had not received CBC training; large classes and inadequate resources hindered implementation	Urban, pre-primary focus; current study targets primary schools in a marginalised ASAL setting
Okeyo and Mokuia (2023)	Preparedness of teachers for CBC in secondary schools, Nyamira South	Schools were ill-prepared; most teachers were untrained; CBAM used	Secondary school focus; current study examines primary schools and measures competence directly
Mutua and Waweru (2023)	Teachers' competence and CBC implementation, Machakos County	Training considered adequate, but materials and high pupil-teacher ratio constrained implementation	Relatively well-served county; current study examines an ASAL context with different constraints
Hussein (2023)	Challenges of CBC implementation in northern Kenya	Poor infrastructure, few resources and limited training undermined implementation	Descriptive study; current study tests the statistical relationship between competence and implementation

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Macheso, Kati, and Wafula (2024)	Use of competency-based assessment tools in western Kenya	Teachers relied on a narrow range of assessment tools	Focused on assessment only; current study includes assessment as part of overall pedagogical competence
Karua et al. (2025)	Teacher preparedness for CBC, Kinangop Sub-County	52.3% of teachers were not competent in preparing schemes of work	Lower primary focus in a highland county; current study covers Grades 1 to 6 in Wajir
Ongubo (2025)	Teacher preparedness for CBC adoption, Garissa County	Over 60% of teachers lacked comprehensive in-service training	Limited to one ward; did not link competence to implementation outcomes
Mwangi et al. (2026)	Teacher training and perceptions prior to CBC implementation	95% trained but 65% held negative perceptions	Did not assess classroom practice; current study uses observation to verify competence
Wawire et al. (2026)	Scoping review of CBC pedagogy and assessment across Africa	Classroom practice remains largely teacher-centred	Regional synthesis; current study provides primary data from a specific marginalised context

Source: Researcher (2026)

From the review, three gaps are evident. There is a contextual gap, since no known study has examined teacher competence and CBC implementation in Habaswein Sub-County or in Wajir County. There is a conceptual gap, because many studies have used attendance at training to represent competence. There is also a methodological gap, since few studies have confirmed self-reported data through classroom observation or tested the separate contribution of each aspect of competence. The present study seeks to fill these gaps.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology that will be used in the study. It presents the research design, location of the study, target population, sampling techniques and sample size, and research instruments. It also describes the pilot study, validity and reliability of the instruments, data collection procedures, data analysis, logistical considerations and ethical considerations.

3.2 Research Design

The study will use a convergent parallel mixed methods design that combines a descriptive survey with a correlational component. In a convergent design, quantitative and qualitative data are collected at the same time and analysed separately, and the results are then merged during interpretation so that one set of results can confirm or explain the other (Creswell & Creswell, 2023). The descriptive survey will enable the researcher to describe the level of teachers' knowledge, their pedagogical competence and the factors affecting their competence without manipulating any variables. The correlational part will enable the researcher to test the strength and direction of the relationship between teachers' competence and effective CBC implementation, as stated in the hypotheses.

This design is suitable because the study seeks both to describe the situation and to test relationships, which requires quantitative data from a relatively large number of teachers. Competence is also a complex concept that may not be fully captured through questionnaires. Interviews with head teachers and Curriculum Support Officers and classroom observation will therefore provide additional evidence. The design has also been used in other Kenyan studies on the CBC. Okeyo and Mokuia (2023) used a convergent mixed methods design to study teacher preparedness in Nyamira South, while Mutua and Waweru (2023) used a descriptive survey to study teacher competence in Machakos. The design requires that the two sets of data are properly integrated, and the researcher will do this by preparing a joint display of the findings during analysis.

3.3 Location of the Study

The study will take place in Habaswein Sub-County, one of the sub-counties of Wajir County in Kenya's North Eastern region. It lies in the southern part of the county, along the main road between Garissa and Wajir towns, and shares a boundary with Garissa County to the south and west. The area is mainly arid, with low and unreliable rainfall,

high temperatures and frequent droughts. Most residents are pastoralists who keep camels, cattle, goats and sheep, and many households move with the seasons in search of water and pasture. Habaswein town is the administrative and trading centre, while the rest of the population lives in small settlements scattered over a very large area (County Government of Wajir, 2023).

Wajir County has some of the lowest education indicators in Kenya. Its County Integrated Development Plan names low enrolment and transition rates, too few teachers, inadequate classrooms and sanitation facilities, and the disruption caused by drought and insecurity as key challenges (County Government of Wajir, 2023). Several public primary schools in the sub-county run low-cost boarding facilities to keep children from mobile households in school. Habaswein was selected because it represents the marginalised ASAL setting in which the CBC is likely to be most difficult to implement, because no previous study has examined teacher competence and CBC implementation in the sub-county, and because the researcher is familiar with the area, which will make it easier to access schools and communicate with respondents.

3.4 Target Population

The target population is made up of all public primary schools in Habaswein Sub-County, their head teachers and teachers, and the Curriculum Support Officers (CSOs) who serve the sub-county. The school will be the unit of analysis, while teachers, head teachers and CSOs will be the units of observation. Teachers are included because they are the implementers of the CBC. Head teachers are included because they are responsible for instructional leadership, supervision of teachers and allocation of resources in their schools. CSOs are included because they provide professional support to teachers and monitor curriculum implementation in schools.

Preliminary information gathered by the researcher suggests that the sub-county has about 42 public primary schools, an estimated 336 teachers (both TSC and Board of Management) and 3 CSOs. These figures are provisional and will be confirmed with the Sub-County Director of Education and the TSC Sub-County Director before data collection starts. Table 3.1 shows how the population is distributed.

Table 3.1: Target Population

Category of School (Stratum)	Number of Schools	Number of Head Teachers	Number of Teachers
Schools in Habaswein town and its environs	12	12	120
Schools in rural settlements (including low-cost boarding schools)	30	30	216

Category of School (Stratum)	Number of Schools	Number of Head Teachers	Number of Teachers
Total	42	42	336

Source: Provisional figures, to be confirmed with the Habaswein Sub-County Education Office (2026)

In addition to the population shown in Table 3.1, the 3 Curriculum Support Officers serving Habaswein Sub-County will form part of the target population.

3.5 Sampling Techniques and Sample Size

3.5.1 Sampling Techniques

The study will use stratified random sampling, simple random sampling, proportionate allocation and census. Schools will first be divided into two strata according to location, that is, schools in Habaswein town and its environs and schools in rural settlements. Stratification is necessary because schools in and around the town are likely to have better access to transport, electricity, training and supervision than rural schools, and these differences may affect teachers' competence. Schools in each stratum will then be selected through simple random sampling using a table of random numbers or a computer-generated list, which gives each school an equal chance of being selected (Creswell & Creswell, 2023).

Fifty percent of the schools in each stratum will be selected, giving 6 schools from the town stratum and 15 from the rural stratum, a total of 21 schools. This proportion is considered adequate for a small population of schools and will allow the researcher to cover both strata within the available time and budget. The head teachers of the 21 sampled schools will automatically be included in the sample. All 3 CSOs will be included through census because their number is small.

3.5.2 Sample Size

The sample size of teachers will be determined using the following formula, which is commonly used in educational surveys where the population is known:

$$n = N / (1 + N(e)^2)$$

Where n is the sample size, N is the target population of teachers (336) and e is the level of precision, set at 0.05 for a 95% confidence level. Substituting these values gives:

$$n = 336 / (1 + 336(0.05)^2) = 336 / 1.84 = 182.6 \approx 183 \text{ teachers}$$

The 183 teachers will be distributed proportionately between the two strata, using the formula: stratum sample = (stratum population / total population) × 183. In each

sampled school, teachers will be listed from the staff register and selected through simple random sampling until the required number for that stratum is reached. Where a school has fewer teachers than required, all its teachers will be included and the balance drawn from other sampled schools in the same stratum. The sample distribution is shown in Table 3.2.

Table 3.2: Sample Size Distribution

Category of Respondents	Target Population	Sample Size	Sampling Technique
Teachers: Habaswein town and environs	120	65	Stratified, proportionate and simple random
Teachers: Rural settlements	216	118	Stratified, proportionate and simple random
Head teachers	42	21	Head teachers of the sampled schools
Curriculum Support Officers	3	3	Census
Total	381	207	

Source: Researcher (2026)

In addition, 21 CBC lessons will be observed, one in each sampled school, using a classroom observation checklist. The teacher to be observed in each school will be randomly selected from the sampled teachers teaching Grades 4 to 6, where most upper primary learning areas and school-based assessments are found. Participation in the observation will be voluntary.

3.6 Research Instruments

Three instruments will be used to collect data, namely a questionnaire for teachers, an interview guide for head teachers and CSOs, and a classroom observation checklist. The use of more than one instrument will make it possible to triangulate the data and improve the credibility of the findings.

3.6.1 Questionnaire for Teachers

The questionnaire for teachers (Appendix III) will be the main instrument for collecting quantitative data. It has six sections. Section A will collect background information, including gender, age, academic and professional qualifications, teaching experience, employer (TSC or Board of Management), grades taught and CBC training attended. Section B will measure knowledge and understanding of the CBC, while Section C will

measure pedagogical competence, including planning, learner-centred methods, integration of competencies and assessment. Section D will cover the factors affecting teachers' competence and Section E the effectiveness of CBC implementation. Section F will have open-ended questions to allow teachers to give their views in their own words.

Sections B to E use five-point Likert items that run from 1 (Strongly Disagree) to 5 (Strongly Agree). Section B also includes a short set of true-or-false knowledge items, so that knowledge is not measured by self-rating alone. A questionnaire is appropriate for this study because it can be used to collect data from many respondents within a short time, it allows anonymity and the data can easily be coded and analysed (Creswell & Creswell, 2023). The questionnaire will be administered in English, which is the language of instruction and the official working language of teachers.

3.6.2 Interview Guide for Head Teachers and Curriculum Support Officers

The 21 head teachers and 3 CSOs will take part in semi-structured interviews based on the guide in Appendix IV. The interviews will seek their views on teachers' knowledge and classroom practice, the support available to teachers, the challenges of implementing the CBC and suggestions for improvement. Semi-structured interviews allow the researcher to probe for more information while keeping the discussion focused on the objectives of the study. Each interview is expected to take between 30 and 45 minutes. With the permission of the interviewee, the interview will be audio-recorded and transcribed later. Where permission is not given, detailed notes will be taken.

3.6.3 Classroom Observation Checklist

A structured observation checklist (Appendix V) will be used to record what takes place during CBC lessons. It will capture evidence of lesson planning, learner-centred methods, the integration of core competencies, values and PCIs, the use of teaching and learning resources, learner participation and assessment. Each item is rated from 1 (Not observed) to 4 (Fully observed), and the observer will also keep short field notes. Observation will help to confirm what teachers report in the questionnaire and will provide direct evidence of implementation. To reduce the effect of the observer's presence, the researcher will enter the classroom a few minutes before the lesson begins and sit at the back without interfering.

3.6.4 Operationalisation of Study Variables

Table 3.3 sets out how each variable will be measured and analysed.

Table 3.3: Operationalisation of Study Variables

Variable	Indicators	Measurement Scale	Instrument	Analysis Technique
Knowledge and understanding of CBC (Independent)	Understanding of goals, core competencies, values, PCIs, learning areas and assessment framework	Ordinal (Likert); knowledge test score	Questionnaire ; interview guide	Frequencies, percentages, means, standard deviation; Pearson correlation; regression
Pedagogical competence (Independent)	Schemes of work and lesson plans; learner-centred methods; integration of competencies; use of resources; competency-based assessment	Ordinal (Likert); observation rating	Questionnaire ; observation checklist	Means, standard deviation; Pearson correlation; regression
Factors affecting competence (Intervening)	Training and CPD; resources and ICT; workload and class size; supervision; attitudes; drought, insecurity and mobility	Ordinal (Likert); open-ended responses	Questionnaire ; interview guide	Frequencies, percentages, means; thematic analysis
Effective CBC implementation (Dependent)	Learner participation; integration of competencies and PCIs; use of varied assessment tools and records; coverage of learning areas; parental involvement	Ordinal (Likert); observation rating	Questionnaire ; observation checklist; interview guide	Means, standard deviation; regression; thematic analysis

Source: Researcher (2026)

3.7 Pilot Study

The instruments will be pre-tested in neighbouring Wajir South Sub-County, which has similar climatic, social and educational conditions to Habaswein but will not be part of the main study. About 10% of the sample will be used in the pilot, that is, 18 teachers, 2 head teachers and 2 classroom observations. The pilot will help the researcher to identify unclear or sensitive items, estimate the time needed to complete each instrument, check the clarity of instructions and test the procedures for coding and data entry. The instruments will be revised based on the results of the pilot before the main data collection.

3.7.1 Validity of the Research Instruments

Validity is the extent to which an instrument measures what it is supposed to measure. The study will focus on content validity and construct validity. For content validity, the instruments will be reviewed by the researcher's supervisors and by two curriculum

studies specialists at Mount Kenya University. Each reviewer will rate how relevant every item is to the study objectives on a four-point scale. A Content Validity Index (CVI) will then be worked out by dividing the number of items rated relevant or highly relevant by the total number of items. An index of 0.78 or more will be taken as acceptable, and items that fall below this mark will be reworded or dropped. Construct validity will be supported by drawing questionnaire items from indicators found in the theories and studies reviewed in Chapter Two, and by checking in the pilot that items meant to measure the same idea are correlated.

3.7.2 Reliability of the Research Instruments

Reliability is the extent to which an instrument gives consistent results under similar conditions. The internal consistency of the Likert sections of the teachers' questionnaire will be checked with Cronbach's alpha, calculated from the pilot data. A coefficient of 0.70 or higher will be regarded as acceptable, in line with common practice in social science research (Field, 2024). If a section falls short, its item-total statistics will be examined and weak items revised or removed. To make the observation checklist more reliable, the researcher and a trained research assistant will observe the same pilot lessons separately and compare their ratings. Where their ratings differ, the differences will be discussed and the checklist descriptors clarified until a high level of agreement is reached.

3.7.3 Trustworthiness of Qualitative Data

The qualitative data will be judged against four criteria: credibility, dependability, confirmability and transferability. Credibility will come from comparing what teachers, head teachers, CSOs and classroom observations reveal, and from member checking, in which some interviewees will be shown a summary of their interview and asked whether it reflects their views accurately. Dependability will be supported by an audit trail recording the decisions made during data collection and analysis. For confirmability, the researcher will keep reflective notes on personal assumptions that might colour the interpretation. Transferability will be helped by a rich description of the study setting, so that readers can decide for themselves whether the findings apply elsewhere (Braun & Clarke, 2022).

3.8 Data Collection Procedures

The researcher will first obtain an introduction letter from the School of Education at Mount Kenya University and ethical approval from the University's Ethics Review Committee. The researcher will then apply for a research permit from the National Commission for Science, Technology and Innovation (NACOSTI). After obtaining the

permit, the researcher will report to the Wajir County Commissioner and the County Director of Education for authority to work in the county, and then to the Sub-County Director of Education and the TSC Sub-County Director in Habaswein. Head teachers of the sampled schools will be contacted beforehand to explain the study and to agree on suitable dates for visits.

Two research assistants will be recruited. They will be qualified teachers who know the area and speak both English and Somali, and they will receive two days of training on the study objectives, the instruments, ethical procedures and personal safety. On arrival at each school, the researcher will report to the head teacher, explain the purpose of the study and seek informed consent from the participants. Questionnaires will be handed to the sampled teachers and, wherever possible, collected the same day to keep the response rate high. Teachers who ask for more time will have up to three days. Head teachers will be interviewed in a private room at the school and CSOs at their offices. Lesson observations will take place at times agreed with the teachers. Data collection is expected to take about six weeks. It will be carried out during the school term, avoiding examination periods and, as far as possible, the peak of the dry season when some families migrate with their livestock.

3.9 Data Analysis and Presentation

3.9.1 Quantitative Data Analysis

After data collection, the completed questionnaires and observation checklists will be checked for completeness and consistency, coded and entered into the Statistical Package for the Social Sciences (SPSS), version 29. The data will then be cleaned by checking for missing values, outliers and entry errors. Frequencies, percentages, means and standard deviations will be used to describe the respondents and to answer the first three research questions. Mean scores on the Likert items will be interpreted as follows: 1.00 to 1.80 very low, 1.81 to 2.60 low, 2.61 to 3.40 moderate, 3.41 to 4.20 high and 4.21 to 5.00 very high.

The hypotheses under the fourth objective will be tested with inferential statistics. Pearson's product-moment correlation will show how strong the relationship is, and in which direction, between each dimension of teachers' competence and effective CBC implementation (H01 and H02). Multiple linear regression will then show the combined and separate influence of the two dimensions of competence on effective implementation (H03). The regression model will be as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Where Y is effective CBC implementation, β_0 is the constant, β_1 and β_2 are the regression coefficients, X_1 is teachers' knowledge and understanding of the CBC, X_2 is teachers' pedagogical competence and ϵ is the error term. Composite scores for each variable will be computed as the mean of the relevant items. Before running the regression, the researcher will check that its assumptions hold, namely normality, linearity, homoscedasticity, independence of errors and the absence of multicollinearity, using histograms, scatter plots, the Durbin-Watson statistic and variance inflation factors (Field, 2024). The coefficient of determination (R^2) will show how much of the variation in implementation is explained by teachers' competence, and the F-test will show whether the model as a whole is significant. All hypotheses will be tested at the 0.05 level of significance. Results will be presented in tables, bar charts and pie charts, each followed by interpretation and discussion.

3.9.2 Qualitative Data Analysis

Data from interviews, open-ended questionnaire items and observation notes will be analysed through reflexive thematic analysis, following the six phases set out by Braun and Clarke (2022). The phases are familiarisation with the data, generating initial codes, constructing themes, reviewing themes, defining and naming themes, and writing the report. Interview recordings will be transcribed word for word. Any answers given in Somali will be translated into English, and a second translator will check the translation. The researcher will read the transcripts several times, code them and group the codes into themes linked to the study objectives, using NVivo to keep the codes and themes organised. The findings will be presented in narrative form and supported by direct quotations. Participants will be identified only by codes, for example HT1 for the first head teacher and CSO2 for the second Curriculum Support Officer.

3.9.3 Integration of Quantitative and Qualitative Findings

In line with the convergent design, the quantitative and qualitative results will be merged during interpretation. For each objective, the researcher will prepare a joint display that sets the main statistical results beside the related qualitative themes and observation evidence. Where the two sets of results agree, this will increase confidence in the findings. Where they differ, for example where teachers rate their competence higher than what is observed in their lessons, the researcher will explore and explain the differences (Creswell & Creswell, 2023).

3.10 Logistical Considerations

Because of the long distances between schools and the security situation in the sub-county, fieldwork will require careful planning. The researcher will hire a reliable

vehicle and plan the visits so that schools in the same area are covered on the same day. Travel will take place only in daylight, and the local administration and security officers will be consulted before visits to remote schools. The research assistants will come from the local community, to make communication easier and build trust with respondents. The work plan and budget appear in Appendices VI and VII.

3.11 Ethical Considerations

The study will follow the ethical standards expected of research with human participants. Ethical approval will be sought from the Mount Kenya University Ethics Review Committee, and a NACOSTI research permit will be obtained before any data are collected. Permission will also be sought from the county and sub-county education authorities and from the head teachers of the sampled schools.

Participation in the study will be voluntary. Every participant will receive an information sheet and consent form (Appendix II) explaining what the study is about, what taking part involves, how long it will take and that they may withdraw at any time without penalty. Written consent will be obtained before any questionnaire is completed, any interview is held or any lesson is observed. Teachers will be assured that the observation is not an appraisal and that nothing seen will be passed to their employer or used for disciplinary purposes.

The confidentiality and anonymity of participants will be protected. Respondents will not write their names on the questionnaires, and schools and individuals will be referred to by codes in every report. Recordings, transcripts and completed instruments will be kept securely, and electronic files will be password-protected in line with the Data Protection Act of 2019. The data will be used only for academic purposes and destroyed once the period required by the university has passed. The researcher will respect the cultural and religious norms of the community, including dress codes, prayer times and gender norms, and will hold interviews at times and places that suit participants. Lastly, all sources used in the study will be properly acknowledged using APA style, and the final report will be checked for plagiarism in line with university policy.

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APPENDICES

The following appendices contain the research instruments and supporting documents for the study.

APPENDIX I: LETTER OF INTRODUCTION

[Full Name of Student]

Mount Kenya University

School of Education

P.O. Box 342-01000

Thika

[Date]

The Head Teacher,

_____ Primary School,

Habaswein Sub-County,

Wajir County.

Dear Sir/Madam,

RE: REQUEST TO COLLECT RESEARCH DATA IN YOUR SCHOOL

I am a postgraduate student at Mount Kenya University pursuing a Master of Education degree in Curriculum Studies. As part of the requirements for the award of the degree, I am carrying out a study titled “Teachers’ Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.”

Your school has been selected to take part in the study. I kindly request your permission to administer questionnaires to some of your teachers, to interview you and to observe one CBC lesson at a time convenient to the school. Participation is voluntary, and the information collected will be treated with strict confidentiality and used for academic purposes only. Neither your name nor the name of your school will appear in any report.

Attached are copies of my introduction letter from the university and the research permit from NACOSTI. Thank you in advance for your cooperation.

Yours faithfully,

[Full Name of Student]

[Registration Number]

APPENDIX II: INFORMED CONSENT FORM

Study title: Teachers' Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.

Purpose of the study: The study seeks to assess teachers' competence in implementing the CBC and to establish how this competence influences the effectiveness of implementation. The findings are expected to inform teacher support programmes and education policy for marginalised areas.

What participation involves: If you agree to take part, you will be asked to complete a questionnaire (about 30 minutes), take part in an interview (about 40 minutes) or allow the researcher to observe one of your lessons, depending on your category of respondent.

Risks and benefits: There are no known risks in taking part in this study. The lesson observation is not an appraisal and will not be shared with your employer. Although you will not receive any payment, your views will contribute to improving teacher support in Habaswein and similar areas.

Confidentiality: Your name will not be recorded on any instrument. All information will be coded, stored securely and used only for this study.

Voluntary participation: Your participation is entirely voluntary. You may decline to answer any question or withdraw at any time without any penalty.

Contact: If you have questions about the study, please contact the researcher on [phone number] or [email address], or the Mount Kenya University Ethics Review Committee.

Declaration by participant

I have read (or have had read to me) the information above. I understand the purpose of the study and what my participation involves, and I voluntarily agree to take part.

Participant's signature: _____ Date: _____

Researcher's signature: _____ Date: _____

APPENDIX III: QUESTIONNAIRE FOR TEACHERS

Instructions

This questionnaire seeks information on teachers' competence and the implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County. Please answer all questions honestly. Do not write your name or the name of your school anywhere on this questionnaire. Tick (✓) the appropriate box or fill in the spaces provided.

Section A: Demographic Information

1. Gender: Male ☐ Female ☐
2. Age bracket (years): Below 30 ☐ 30–39 ☐ 40–49 ☐ 50 and above ☐
3. Highest professional qualification: P1 Certificate ☐ Diploma ☐ Bachelor's degree ☐ Master's degree ☐ Other (specify) _____
4. Employer: TSC ☐ Board of Management ☐ Other (specify) _____
5. Teaching experience (years): Below 5 ☐ 5–10 ☐ 11–15 ☐ Above 15 ☐
6. Grade(s) currently taught: Grade 1–3 ☐ Grade 4–6 ☐ Both ☐
7. Have you attended any CBC training? Yes ☐ No ☐
8. If yes, how many days of CBC training have you attended in total? 1–5 days ☐ 6–10 days ☐ More than 10 days ☐
9. Location of your school: Habaswein town and environs ☐ Rural settlement ☐

Section B: Teachers' Knowledge and Understanding of the CBC

Part I: Indicate the extent to which you agree with each statement. Key: SD = Strongly Disagree (1); D = Disagree (2); N = Neutral (3); A = Agree (4); SA = Strongly Agree (5).

No.	Statement	SD	D	N	A	SA
B1	I clearly understand the vision and mission of the Competency-Based Curriculum.					
B2	I can explain each of the seven core competencies of the CBC.					
B3	I understand how national values should be integrated into lessons.					
B4	I can identify pertinent and contemporary issues (PCIs) relevant to the learning areas I teach.					

No.	Statement	SD	D	N	A	SA
B5	I understand the learning areas and their specific learning outcomes for the grades I teach.					
B6	I understand the KNEC competency-based assessment framework and the four performance levels.					
B7	I understand the role of parents in supporting learning under the CBC.					
B8	I can interpret the curriculum designs for the learning areas I teach without difficulty.					

Part II: For each statement, tick True or False.

No.	Statement	True	False
B9	Under the CBC, the teacher's main role is to act as a facilitator of learning.		
B10	Learner performance in CBC is reported using four levels of achievement.		
B11	Core competencies should only be taught in specific learning areas such as Science and Technology.		
B12	Portfolios, rubrics and observation schedules are examples of CBC assessment tools.		
B13	Summative examinations are the only valid way to assess learners under the CBC.		

Section C: Teachers' Pedagogical Competence

Indicate the extent to which you agree with each statement. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
C1	I prepare schemes of work in the format required for the CBC.					
C2	I prepare lesson plans that clearly state specific learning outcomes and key inquiry questions.					
C3	I regularly use group work, discussion and cooperative learning in my lessons.					
C4	I use inquiry, projects and problem-solving activities to help learners discover knowledge.					
C5	I deliberately integrate core competencies and values into my lessons.					
C6	I improvise teaching and learning resources from locally available materials.					

No.	Statement	SD	D	N	A	SA
C7	I adapt my teaching to meet the needs of learners with different abilities.					
C8	I use a variety of assessment tools such as observation, checklists, rubrics and portfolios.					
C9	I keep up-to-date assessment records for every learner in my class.					
C10	I give learners feedback that helps them to improve their learning.					
C11	I integrate digital or low-cost technology into my lessons where possible.					

Section D: Factors Affecting Teachers' Competence in Implementing the CBC

Indicate the extent to which you agree that each of the following affects your competence in implementing the CBC. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
D1	The CBC training I received was adequate to prepare me to implement the curriculum.					
D2	I have regular opportunities for continuous professional development on the CBC.					
D3	My school has adequate textbooks, curriculum designs and teaching materials.					
D4	My school has adequate ICT facilities, electricity and internet to support CBC teaching.					
D5	My workload allows me enough time to plan and assess learners as required by the CBC.					
D6	The number of learners in my class allows me to use learner-centred methods effectively.					
D7	The head teacher regularly supervises and supports my CBC teaching.					
D8	Curriculum Support Officers visit and support teachers in my school.					
D9	I have a positive attitude towards the CBC.					
D10	Drought and the movement of pastoralist families disrupt my teaching and assessment.					
D11	Insecurity in the area affects my ability to attend training and deliver the curriculum.					
D12	Parents in this community support their children's learning activities under the CBC.					

Section E: Effective Implementation of the CBC

Indicate the extent to which you agree with each statement about CBC implementation in your class. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
E1	Learners actively participate in activities during my lessons.					
E2	Learners in my class demonstrate core competencies such as communication and collaboration.					
E3	PCIs and values are integrated in most of my lessons.					
E4	I use a variety of formative assessment tools as required by the CBC.					
E5	I cover the learning areas in the time allocated in the curriculum designs.					
E6	Learners complete projects and practical activities as required.					
E7	Parents participate in their children's learning activities when asked.					
E8	Overall, the CBC is being implemented effectively in my class.					

Section F: General Comments

1. In your opinion, what are the main challenges you face in implementing the CBC in this sub-county?

2. What kind of support would most improve your competence in implementing the CBC?

3. Any other comments on CBC implementation in Habaswein Sub-County?

Thank you for your time and cooperation.

APPENDIX IV: INTERVIEW GUIDE FOR HEAD TEACHERS AND CURRICULUM SUPPORT OFFICERS

Introduction: Thank you for agreeing to take part in this interview. The purpose is to understand your views on teachers' competence and the implementation of the CBC in Habaswein Sub-County. The interview will take about 40 minutes. With your permission, it will be recorded. Your responses will remain confidential.

Part A: Background

1. How long have you served as a head teacher or Curriculum Support Officer, and how long have you been in Habaswein Sub-County?
2. What CBC training have you personally received?

Part B: Teachers' Knowledge and Understanding of the CBC

3. How would you describe the level of understanding of the CBC among teachers in your school or zone?
4. Which aspects of the curriculum do teachers seem to understand well, and which do they find most difficult?

Part C: Teachers' Pedagogical Competence

5. How do teachers in your school or zone plan and deliver CBC lessons? What methods do they commonly use?
6. How well do teachers carry out competency-based assessment and keep learner records?
7. Can you give an example of a lesson you observed that reflected the CBC very well, or one that did not?

Part D: Factors Affecting Teachers' Competence

8. What training and professional development opportunities are available to teachers here? How adequate are they?
9. How do resources, staffing and class sizes affect teachers' ability to implement the CBC?
10. In what ways do local conditions such as drought, pastoralist mobility and insecurity affect teaching and learning?
11. What support do you provide to teachers, and what challenges do you face in providing it?

Part E: Competence and Effective Implementation

12. In your experience, how does the competence of a teacher affect the way the CBC is implemented in the classroom?
13. What differences do you notice between schools or classes where the CBC is implemented well and those where it is not?
14. What should the Ministry of Education, TSC, KICD and the County Government do to strengthen teachers' competence in this sub-county?

Thank you for your time and valuable contribution.

APPENDIX V: CLASSROOM OBSERVATION CHECKLIST

School code: _____ Grade: _____ Learning area: _____ Date: _____

Number of learners present: _____ Duration of lesson: _____

Rating key: 1 = Not observed; 2 = Partly observed; 3 = Largely observed; 4 = Fully observed

No.	Aspect Observed	1	2	3	4
1	Teacher has a lesson plan in the CBC format with clear learning outcomes				
2	Lesson begins with a key inquiry question or activity that connects to learners' experience				
3	Learners work in pairs or groups during the lesson				
4	Learners are engaged in inquiry, problem solving or hands-on activities				
5	Teacher acts as a facilitator rather than the main source of information				
6	At least one core competency is deliberately developed				
7	Values and/or PCIs are integrated into the lesson				
8	Teacher uses relevant teaching and learning resources, including improvised materials				
9	Technology (digital or low-cost) is used appropriately				
10	Teacher uses formative assessment during the lesson (questions, observation, checklists)				
11	Learners receive feedback on their work				
12	Learners with different abilities are supported				
13	Assessment records and learner portfolios are available and up to date				
14	Lesson concludes with a reflection or summary involving learners				

Observer's field notes:

APPENDIX VI: WORK PLAN

Table A.1: Work Plan, October 2026 – September 2027

Activity	Oct– Nov 2026	Dec 2026– Jan 2027	Feb– Mar 2027	Apr– May 2027	Jun– Jul 2027	Aug– Sep 2027
Proposal correction and defence	✓					
Ethical approval and NACOSTI permit		✓				
Pilot study and revision of instruments		✓				
Main data collection			✓			
Data entry, cleaning and analysis				✓		
Writing of Chapters Four and Five				✓	✓	
Submission of draft thesis to supervisors					✓	
Corrections, final defence and submission						✓

APPENDIX VII: RESEARCH BUDGET

Table A.2: Research Budget

Item	Quantity	Unit Cost (KES)	Total Cost (KES)
Stationery, printing and binding of proposal	6 copies	1,500	9,000
Printing of questionnaires and other instruments	250 sets	60	15,000
NACOSTI research permit	1	1,000	1,000
Training of research assistants	2 persons × 2 days	2,500	10,000
Research assistants' allowance (fieldwork)	2 persons × 30 days	1,500	90,000
Vehicle hire and fuel for fieldwork	30 days	6,000	180,000
Accommodation and meals for researcher	30 days	3,000	90,000
Pilot study expenses	Lump sum	20,000	20,000
Airtime and internet	Lump sum	8,000	8,000
Transcription and translation of interviews	24 interviews	1,000	24,000
Data analysis software and services	Lump sum	25,000	25,000
Printing and binding of final thesis	6 copies	2,000	12,000
Subtotal			484,000
Contingency (10%)			48,400
Grand Total			532,400